

Exercise 6.1

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{3!} \phi^3 \quad \text{in six space-time dimensions.}$$

The Wilsonian approach leads us to the effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{1+\kappa}{2} (\partial_\mu \phi \partial^\mu \phi) + \frac{1}{2} (m^2 + \mu^2) \phi^2 + \frac{\lambda + \xi}{3!} \phi^3 + \tau \phi + \dots,$$

where ϕ is a field with Fourier modes $|k| < b\Lambda$.

Our first task is to find expression for κ, μ^2, ξ (using ξ instead of ζ since it's easier to write)
 (We will write the Feynman-rules after doing the integrations)
 the corrections κ and μ^2 are corrections to the propagator
 while ξ is correction of 3-point function. Let's first calculate the correction ξ

$$\xi \sim \text{diagram} \sim \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)((k-p)^2 + m^2)((k-p-q)^2 + m^2)}$$

$b\Lambda \leq |k| \leq \Lambda$

we take the limit $m^2 \ll \Lambda^2$ ^{mass is perturbation} and low momentum modes external legs
 $\Rightarrow p \ll \Lambda, q \ll \Lambda$. So we can approximate the integral to

$$J \equiv \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^6} = \int \frac{d\Omega_D}{(2\pi)^D} \int_{b\Lambda}^{\Lambda} dk \frac{1}{k^6} \quad \left\| \int d\Omega_D = \frac{2\pi^3}{\Gamma(6/2)} = \pi^3 \right.$$

$D=6$, need to take $D=6-\epsilon$ limit

$$= \frac{2^3 \pi^3}{2^3 (2\pi)^6} \left[\ln \Lambda - \ln(b\Lambda) \right] = \frac{1}{8 (2\pi)^3} \ln\left(\frac{1}{b}\right)$$

$$J = \frac{1}{8 (2\pi)^3} \ln(1/b)$$

Now let's look at the one-loop correction of the propagator:

$$p \text{ --- } \text{diagram} \text{ --- } p-k \sim \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)((k-p)^2 + m^2)} = I \quad \left\| \text{Let's now use Feynman Parametrization} \right.$$

$$\Rightarrow I = \int_0^1 dx \int \frac{d^D k}{(2\pi)^D} \frac{1}{[x(k^2 + m^2) + (1-x)((k-p)^2 + m^2)]^2}$$

Let's simplify the denominator: $xk^2 + x p^2 + k^2 - 2kp + p^2 - xk^2 + 2kpx - xp^2 + m^2 - xp^2 = \mathcal{F}$

$$\Rightarrow \mathcal{F} = k^2 - 2kp(1-x) + [(1-x)p]^2 - [(1-x)p]^2 + p^2 - xp^2 + m^2$$

let's now do a shift $l = k - (1-x)p \rightarrow dl = dk$

and write $\Delta(p^2) = -(1-x)^2 p^2 + (1-x)p^2 + m^2$

$\Rightarrow \mathcal{L} = l^2 + \Delta(p^2)$ (We are working in Euclidean space)

$\Rightarrow I = \int_0^1 dx \int \frac{d^D l}{(2\pi)^D} \frac{1}{(l^2 + \Delta(p^2))^2} = \int_0^1 dx \int \frac{d^D l}{(2\pi)^D} \frac{1}{l^4} \left(1 + \frac{\Delta(p^2)}{l^2}\right)^{-2}$

Let's now take $p \ll \Lambda$ and $m \ll \Lambda$ and expand in $\Delta(p^2)$

such that $\left(1 + \frac{\Delta(p^2)}{l^2}\right)^{-2} \approx 1 - 2 \frac{\Delta(p^2)}{l^2}$

$\Rightarrow I = \int_0^1 dx \int \frac{d^D l}{(2\pi)^D} \frac{1}{l^4} \left(1 - 2 \frac{\Delta(p^2)}{l^2}\right)$ now let's go to

spherical coordinates and take $D=6$ so that $\int d\Omega_6 = \pi^3$

$I = \frac{\pi^3}{(2\pi)^6} \int_0^1 dx \int_{b\Lambda}^{\Lambda} dl \left(l - 2 \Delta(p^2) \frac{1}{l} \right)$

$I = \frac{1}{8(2\pi)^3} \left[\frac{1}{2} \Lambda^2 (1-b^2) - \int_0^1 dx 2 \Delta(p^2) \ln(1/b) \right]$

Since $\Lambda \gg p$ we can set the limit of l -integral to be same as that of k -integral.

Also we can drop the m^2 -term from $\Delta(p^2)$ since the contribution of term proportional is negligible for contribution

to correction of mass since $\Lambda \gg m^2$ and we have $\frac{1}{2} \Lambda^2 (1-b^2) - 2m^2 \ln(1/b) \approx \frac{1}{2} \Lambda^2 (1-b^2)$

from $\Delta(p^2) = -(1+x)^2 p^2 + (1-x)p^2 + m^2$ we neglect the m^2 term for following reason $\rightarrow$

$\Rightarrow \Delta(p^2) \approx (1-x)x p^2 \Rightarrow \int_0^1 dx (x-x^2) p^2 = \left(\frac{1}{2} - \frac{1}{3}\right) p^2 = \frac{1}{6} p^2$

$\Rightarrow I = \frac{1}{8(2\pi)^3} \left[\frac{1}{2} \Lambda^2 (1-b^2) - \frac{2}{6} p^2 \ln(1/b) \right]$

the term proportional to p^2 gives correction to $(\partial\phi)^2$ (since $\partial_\mu \phi \rightarrow p$ with Fourier transform) and the other term gives correction to mass

Now we need to get the correct coefficients to actually get the correct expression for μ^2 , κ and ξ

Let's now write the Feynman rules for Lagrangian before integrating out high momentum modes: $\mathcal{L} = (\partial\phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{3!} \phi^3 + (\partial\hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 + \frac{\lambda}{3!} \hat{\phi}^3 + (3\phi\hat{\phi}\hat{\phi} + 3\phi^2\hat{\phi}) \frac{\lambda}{3!}$

From this we get Feynman rules

$\phi\phi\hat{\phi}: \text{---} \times \text{---} = -\frac{\lambda}{2}$

and $\phi\hat{\phi}\hat{\phi}: \text{---} \times \text{---} = -\frac{\lambda}{2}$

and $\text{---} = \frac{1}{k^2 + m^2} \Theta(k)$ and also $\text{---} = -\frac{\lambda}{3!}$ for only ϕ terms (no $\hat{\phi}$)
 with $\Theta(k) = \begin{cases} 1 & \text{if } b \leq |k| \leq a \\ 0 & \text{elsewhere} \end{cases}$ with the other Feynman rules $\hat{\phi}$ being standard Feynman rules for ϕ^3 -theory eg $\text{---} = -\lambda$

For our task we will only be needing --- vertex Feynman rules.

Now let's find μ^2, k_0, ξ :

To do this let's follow the approach of Peskin & Schroeder eq (12.10) onwards

Our two point function correction comes from $\int d^D x \left(\frac{\lambda}{2}\right)^2 \phi^2 \hat{\phi} \hat{\phi}$
 coeff of ϕ^2 and ϕ^2 term in Lagrangian two 3-vertex: ---

$$\text{and } \int d^D x \left(\frac{\lambda}{2}\right)^2 \phi^2 \hat{\phi} \hat{\phi} = -\frac{1}{2} \int \frac{d^D k}{(2\pi)^D} [\mu^2 + k p^2] \phi(k) \phi(-k) \quad (1)$$

$$\text{Such that } \mu^2 + k p^2 = -\frac{\lambda^2}{2} \cdot \frac{1}{2!} \cdot 2 \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)((k-p)^2 + m^2)} \quad (2)$$

$= I$

$\mu^2 + k p^2 = -\frac{\lambda^2}{2!} I$, now let's explain all the multiplication coefficients from (2): $\lambda^2/2$ comes from eq. (1)'s relation $\frac{1}{2!}$ comes from the fact that the exponential of the action

$\exp(-\int d^D x \mathcal{L}_{\text{eff}})$ since $\mu^2 + k p^2$ are in order λ^2 so the $\frac{1}{2!}$ factor, for expansion of exponential to the second order in λ . The term 2

comes from symmetry factor of the loop: ---

The $-$ sign for $(\mu^2 + k p^2)$ also comes from the term being of order λ^2 in expansion which keeps $\int d^D x \left(\frac{\lambda}{2}\right)^2 \phi^2 \hat{\phi} \hat{\phi}$ from having minus sign like in Peskin (12.10)

$$\text{So we get } \mu^2 + k p^2 = -\frac{1}{2} I = -\frac{\lambda^2}{32(2\pi)^3} \Lambda^2 (1-b^2) + \frac{\lambda^2}{48(2\pi)^3} p^2 \ln(1/b)$$

$$\text{We see that } \mu^2 = -\frac{\lambda^2}{32(2\pi)^3} \Lambda^2 (1-b^2) \quad \square \quad \text{and} \quad k_0 = \frac{\lambda^2}{48(2\pi)^3} \ln(1/b) \quad \square$$

Which are the coefficients we needed to derive. Now let's do some analysis for ξ : $\xi \sim \lambda^3 \Rightarrow$ is the λ^3 -expansion of exponential $\Rightarrow \frac{1}{3!}$ -term will appear. Now let's start with

minus comes since $\Rightarrow (-\lambda)^3$ -term

$$-\int d^D x \left(\frac{\lambda}{2}\right)^3 \phi^3 \hat{\phi} \hat{\phi} \hat{\phi} = -\frac{1}{3!} \int d^D x \xi \phi^3$$

coeff of ϕ^3 term in Lagrangian

Not sure if it should actually be $\frac{d^D k}{(2\pi)^D}$ like in previous case but since

I am following Peskin and for ϕ^4 he has $d^D x$ in eq. (12.14) I will follow his convention.

$$\text{Now we get } \xi = 3! \left(\frac{\lambda}{2}\right)^3 \frac{2}{3!} \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^6}$$

$$\xi = \frac{\lambda^3}{4} J = \frac{\lambda^3}{32(2\pi)^3} \ln(1/b)$$

Here once again we get $\frac{1}{3!}$ term from expansion of exponential and $3! \left(\frac{\lambda}{2}\right)^3$ from matching $-\int d^D x \phi^3 \hat{\phi} \hat{\phi} \hat{\phi} \left(\frac{\lambda}{2}\right)^3 = -\frac{1}{3!} \int d^D x \xi \phi^3$.

$$\Rightarrow \underline{\underline{\xi = \zeta = \frac{\lambda^3}{32(2\pi)^3} \ln(1/b)}}$$

by rescaling field such that $\phi' = (1+\kappa)^{1/2} \phi$ we get effective coupling from $\frac{1}{3!} \lambda' \phi'^3$ term in rescaled Lagrangian

$$\rightarrow \frac{1}{3!} \lambda' \phi'^3 = \frac{\lambda + \xi}{3!} \phi^3 \Rightarrow \lambda' = \frac{\lambda + \xi}{(1+\kappa)^{3/2}}$$

Now expand λ' in λ with $\lambda^2 \ll 1$

$$\lambda' = \frac{\lambda + \frac{\lambda^3}{32(2\pi)^3} \ln(1/b)}{\left[1 + \frac{\lambda^2}{32(2\pi)^3} \ln(1/b)\right]^{3/2}} \approx \left[\lambda + \frac{\lambda^3}{32(2\pi)^3} \ln(1/b)\right] \left[1 - \frac{3}{2} \frac{\lambda^2}{32(2\pi)^3} \ln(1/b)\right]$$

$$\Rightarrow \lambda' = \lambda - \frac{3}{2} \lambda^2 \frac{1}{32(2\pi)^3} \ln(1/b) + \frac{\lambda^3}{32(2\pi)^3} \ln(1/b) - \frac{3}{2} \lambda^5 \left[\frac{1}{32(2\pi)^3}\right]^2 \left(\ln(1/b)\right)^2$$

We can see that the coupling diverges for low-momentum ($b \rightarrow 0$) at 1-loop order (at 1-loop order we have $\mathcal{O}(\lambda^2)$ -terms in addition to λ)

$$\Rightarrow \lambda' = \frac{1 + \xi}{(1+\kappa)^{3/2}} \text{ implies } \uparrow \text{ one-loop running of the coupling in } \phi^3\text{-theory.}$$

Exercise 6.2

$$\mu \frac{d}{d\mu} \left[\langle \Omega | T(\phi_0(x_1) \dots \phi_0(x_n)) | \Omega \rangle \right] = 0 \quad (1)$$

We define $G^{(n)}(x_1, \dots, x_n) = \langle \Omega | T(\phi(x_1) \dots \phi(x_n)) | \Omega \rangle$
where $\phi_0 = \sqrt{z} \phi$

$$\Rightarrow G^{(n)} = z^{-n/2} \langle \Omega | T(\phi_0(x_1) \dots \phi_0(x_n)) | \Omega \rangle$$

now let's go back to eq. (1) in terms of $G^{(n)}$

$$\Rightarrow \mu \frac{d}{d\mu} \left(z^{+n/2} G^{(n)}(x_1, \dots, x_n) \right) = 0$$

Now we can make use of derivative of products: $\frac{d}{dx} (y \cdot z) = y \frac{dz}{dx} + z \frac{dy}{dx}$

$$\Rightarrow \mu \frac{d}{d\mu} (z^{+n/2} G^{(n)}(x_1, \dots, x_n)) = \mu \frac{d}{d\mu} (z^{+n/2}) G^{(n)}(x_1, \dots, x_n) + z^{+n/2} \mu \frac{d}{d\mu} G^{(n)}(x_1, \dots, x_n) = 0$$

Now we know that G also depends on coupling (g), mass (m) and scale μ : $G^{(n)}(x_1, \dots, x_n, g, m, \mu)$ So we can write the total differential in terms of partial differential such that

$$\frac{d}{d\mu} G^{(n)}(x_1, \dots, x_n, g, m, \mu) = \left(\frac{\partial}{\partial \mu} + \frac{\partial g}{\partial \mu} \frac{\partial}{\partial g} + \frac{\partial m}{\partial \mu} \frac{\partial}{\partial m} \right) G^{(n)}(x_1, \dots, x_n, g, m, \mu)$$

Let's write $x_1, \dots, x_n = x_i$

$\Rightarrow$ So our original equation (1) is now

$$\left[\mu \frac{dz^{+n/2}}{d\mu} + z^{+n/2} \left(\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} - \gamma_m(g, m) m \frac{\partial}{\partial m} \right) \right] G^{(n)}(x_i, g, \mu, m) = 0$$

with $\beta = \mu \frac{\partial g}{\partial \mu}$, $\gamma_m(g, m) = -\frac{\mu}{m} \frac{\partial m}{\partial \mu}$

Now let's look at $\mu \frac{dz^{+n/2}}{d\mu}$ term:

$$\mu \frac{dz^{+n/2}}{d\mu} = \mu (+n/2) z^{+n/2-1} \frac{dz}{d\mu} = + z^{+n/2} n \frac{\mu}{z} \frac{dz}{d\mu}$$

Let's now define $\gamma_\phi(g, m) \equiv \frac{\mu}{2z} \frac{dz}{d\mu} \Rightarrow \mu \frac{dz^{+n/2}}{d\mu} = z^{+n/2} n \gamma_\phi(g, m)$

$\Rightarrow$ Our equation from above becomes now:

$$\left[z^{+n/2} n \gamma_\phi(g, m) + z^{+n/2} \left[\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} - \gamma_m(g, m) m \frac{\partial}{\partial m} \right] \right] G^{(n)}(x_i, g, \mu, m) = 0$$

Now we can see that for this to be true for all $z^{+n/2}$ we need

$$\left[\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} - \gamma_m(g, m) m \frac{\partial}{\partial m} + n \gamma_\phi \right] G^{(n)}(x_i, g, \mu, m) = 0$$

And switching $G^{(n)}$ into momentum space such that $x_i \rightarrow p_i$ we get the desired equation:

$$\left[\mu \frac{\partial}{\partial \mu} + \beta \frac{\partial}{\partial g} - \gamma_m(g, m) m \frac{\partial}{\partial m} + n \gamma_\phi \right] G^{(n)}(p_i, g, \mu, m) = 0$$

Exercise 6.3

Solve
$$\left[\mu \frac{\partial}{\partial \mu} + \beta(g, m) \frac{\partial}{\partial g} - \gamma_m(g, m) m \frac{\partial}{\partial m} + \gamma_\phi n \right] G^{(n)}(p_i; g, \mu, m) = 0$$

Since G depends on external momenta $\{p_i\}$ we may write

$G^{(n)}(p_i; p_i; g, \mu, m)$, let's now rescale this to get the dimensionless G

$G^{(n)}(p_i; p_i; g, \mu, m) = \mu^{-w} G^{(n)}\left(\frac{p_i \cdot p_j}{\mu^2}, \frac{m^2}{\mu^2}, g(\mu^2)\right)$, where $-w$ is the mass dimension of $G^{(n)}$. Let's also rescale $p_i \rightarrow e^{-t} p_i$

$\Rightarrow \mu^{-w} G^{(n)}\left(\frac{p_i \cdot p_j}{e^{2t} \mu^2}, \frac{m^2}{\mu^2}, g(\mu^2)\right) \equiv G_R$

G_R follows the following equation:
$$\left[\mu \frac{\partial}{\partial \mu} \Big|_{g, m, t} - \frac{\partial}{\partial t} \Big|_{g, m, \mu} + m \frac{\partial}{\partial m} \Big|_{g, \mu, t} + w \right] G_R = 0$$

Since $\mu \frac{\partial}{\partial \mu} \Big|_{g, m, t} G_R = -w G_R$ $\Rightarrow$ here we can think of μ -dependence only being in μ^{-w} term, alternatively we get same result by using $\mu \frac{\partial}{\partial \mu} G^{(n)}(p_i; p_i; g, \mu, m)$ and the dimensionality of $G^{(n)}$ for what power of μ is in $G^{(n)}$

$\frac{\partial}{\partial t} \Big|_{g, m, \mu} G_R = -2 G_R$

$m \frac{\partial}{\partial m} \Big|_{g, \mu, t} G_R = -2 \frac{1}{m} G_R = 2 G_R$

$\Rightarrow$ This confirms the equation for G_R , now we put this into the original equation to be solved:

$\Rightarrow \left[\beta(g) \frac{\partial}{\partial g} - (\gamma_m + 1) m \frac{\partial}{\partial m} + \frac{\partial}{\partial t} - w + \gamma_\phi n \right] G_R = 0$ $\left\| \begin{array}{l} \frac{\partial}{\partial g} \equiv \frac{\partial}{\partial g} \Big|_{m, t} \\ \text{so we keep other variables constant} \end{array} \right.$

Let's now define
$$\begin{cases} \frac{\partial \bar{g}(g, t)}{\partial t} = \beta(\bar{g}(g, t)) \\ \bar{g}(g, 0) = g \end{cases} \text{ and } \begin{cases} \frac{\partial \bar{m}(g, t)}{\partial t} = -(1 + \gamma_m(\bar{g}(g, t))) \bar{m}(g, t) \\ \bar{m}(g, 0) = m \end{cases}$$

and write the equation in terms of $\bar{g}$ and $\bar{m}$, also $\gamma_\phi(g) \rightarrow \gamma_\phi(\bar{g})$

$\Rightarrow \left[\frac{\partial}{\partial t} + \beta(\bar{g}) \frac{\partial}{\partial \bar{g}} - (1 + \gamma_m(\bar{g})) \frac{\partial}{\partial \bar{m}} - w + \gamma_\phi(\bar{g}) n \right] \mu^{-w} G^{(n)}\left(\frac{p_i \cdot p_j}{e^{2t} \mu^2}, \frac{\bar{m}(t)}{\mu}, \bar{g}(t)\right) = 0$ $\left\| \begin{array}{l} \text{Now we use chain-rule and definitions of } \beta(\bar{g}), -(1 + \gamma_m(\bar{g})) \bar{m} \text{ to get total differential of } t \\ \int_{t=0}^t \frac{1}{F(t)} dF = \ln(F(t)) - \ln(F(0)) \end{array} \right.$

Can divide μ^{-w} out

$$\left[\frac{d}{dt} - w + n \gamma_\phi(t) \right] G^{(n)}\left(\frac{p_i \cdot p_j}{e^{2t} \mu^2}, \frac{\bar{m}(t)}{\mu}, \bar{g}(t)\right) = 0$$

Now we can integrate this:
$$\int_{G^{(n)}(at, t=0)}^{G^{(n)}(at, t)} \frac{1}{G^{(n)}(t)} dG_R = \int_0^t (w - n \gamma_\phi(t')) dt'$$

We have $\bar{m}(\bar{g}, 0) = m$, and $\bar{g}(\bar{g}, 0) = g$

$\Rightarrow G^{(n)}\left(\frac{p_i \cdot p_j}{e^{2t} \mu^2}, \frac{\bar{m}}{\mu}, \bar{g}\right) = G^{(n)}\left(\frac{p_i \cdot p_j}{e^{2t} \mu^2}, \frac{\bar{m}(t)}{\mu}, \bar{g}(t)\right) \exp\left[-tw + n \int_0^t \gamma_\phi(t') dt'\right]$

We got the solution!